PowerPoint Presentation by

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MANAGEMENT ACCOUNTING

8^{th} EDITION

BY

HANSEN & MOWEN

7 SUPPORT-DEPARTMENT COST ALLOCATION

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

LEARNING OBJECTIVES

1. Describe the difference between support departments and producing departments.
2. Calculate single & multiple charging rates for a support department.
3. Allocate support-department costs to producing departments using the direct, sequential, & reciprocal methods.

Continued

LEARNING OBJECTIVES

4. Compute departmental overhead rates.
5. Describe the allocation of joint costs to products. *(Appendix)*.

Click the button to skip Questions to Think About

QUESTIONS TO THINK ABOUT:
Hamilton & Barry, CPAs

Why do you think that the photocopying charges amount to \$0.12 per page? List types of costs incurred & divide them into fixed & variable categories.

QUESTIONS TO THINK ABOUT:
Hamilton & Barry, CPAs

Jan mentioned the security & convenience of in-house photocopying. How to you think the firm might weigh these factors in deciding whether cost of in-house copying is “worth it”?

QUESTIONS TO THINK ABOUT:
Hamilton & Barry, CPAs

Since the firm as a whole has decided to have an in-house copying department, why are copying costs charged to the individual departments? What purpose does developing support-department charging rates serve?

LEARNING OBJECTIVE

1

Describe the difference between support departments and producing departments.

ALLOCATION: Definition

A means of dividing a pool of costs & assigning it to various subunits.

COST ALLOCATION

While cost allocation does not affect total product cost, it will affect pricing & profitability of individual products depending on method used.

COMMON COSTS: Definition

Mutually beneficial costs which occur when the same resource is used in output of 2 or more services or products.

TYPES OF DEPARTMENTS

Producing departments are directly responsible for creating products, services sold. Support departments provide essential support services for producing departments.

How are overhead costs treated for producing & support departments?

Once producing & support departments are identified, overhead costs are traced, not allocated to each department.

Why can’t a support department have an overhead rate to assign to products?

Support departments do not produce salable products.

CAUSAL FACTORS: Definition

Activities within a producing department that provoke the incurrence of support service costs.

TYPES OF DEPARTMENTS:
Examples

- Manufacturing plant
  - Producing departments (Assembly & Finishing)
  - Support departments (Storeroom, Cafeteria, Maintenance, General Factory)
- Bank
  - Producing (Auto Loans, Commercial Lending, Personal Banking
  - Support departments (Drive-Thru, Data Processing, Bank Administration)

How are costs allocated from departments to products?

First, support department costs are assigned to producing departments. Then overhead rates are developed to cost products.

OBJECTIVES OF ALLOCATION

- To obtain a mutually agreeable price
- To compute product-line profitability
- To predict the economic effects of planning & control
- To value inventory
- To motivate managers

LO 1

COMPETITIVE PRICING

- Requires understanding costs
  - Overstating leads to loss of business
  - Understating produces losses
- Leads to evaluating product or service mix
  - Dropping some services
  - Reallocating resources
  - Repricing

LEARNING OBJECTIVE

2

Calculate single & multiple charging rates for a support department.

What kinds of charging rates are used?

Companies use either a **single** charging rate or **multiple** charging rates.

PHOTOCOPYING DEPT.:
Barry & Hamilton

Service department usage	
Audit department	94,500
Tax department	67,500
MAS department	108,000
Total	270,000
Costs	
Fixed	$ 26,190
Estimated variable	6,210

FORMULAS: Single Charging Rate

Charging rate =

Total estimated costs / Estimated usage

$ 32,400 / 270,000 = $0.12 per page

Allocating charges:

 # Pages *x* Charging rate = Allocated charges

CHARGE ALLOCATION: Single Charging Rate

Dept	# Pages	Charge Rate	Total Charges
Audit	92,000	$ 0.12	$ 11,040
Tax	65,000	0.12	7,800
MAS	115,000	0.12	13,800
Total	272,000		$ 32,640

What do you need to know to use multiple charging rates?

Multiple charging rates require that causal factors are known.

PHOTOCOPYING DEPT: Causal Factors

Causal factor for size & costs of photocopying is monthly peak usage.

FORMULAS: Multiple Charging Rates

Peak usage =	
Average usage Audit + MAS	*16875*
Peak usage, Tax	*22,500*
Peak usage	*39,375*

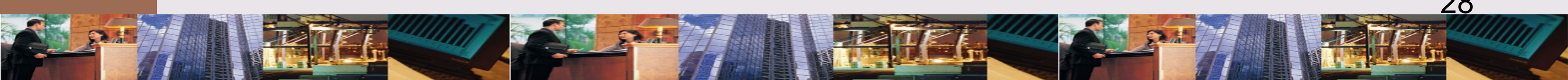
Allocating charges:

Fixed costs = Proportion Peak x Fixed Cost

Variable costs = Estimated Variable cost *x* Actual page usage

FIXED COST ALLOCATION

Dept	Peak # Pages	Proportion Peak Usage	Total Fixed Cost	Total Charges
Audit	7,875	0.20	$ 26,190	$ 5,238
Tax	22,500	0.57	26,190	14,928
MAS	9,000	0.23	26,190	6,024
Total	**39,375**			$ **26,190**



COST ALLOCATION: Multiple Charging Rates

Dept	Total # Pages	Variable Cost @ $0.023	Fixed Cost Allocation	Total Charges
Audit	92,000	$ 2,116	$ 5,238	$ 7,354
Tax	65,000	1,495	14,928	16,423
MAS	115,000	2,645	6,024	8,669
Total	272,000	$ 6256	$ 26,190	$ 32,446

What are the uses of **budgeted costs**?

Budgeted costs are used 1) to help determine overhead rate and 2) for service department performance evaluation .

PERFORMANCE EVALUATION

- General principle
  - Managers should not be held responsible for cost or activities over which they have no control
- Corollary
  - Actual costs should not be allocated to producing departments because they include either <u>*efficiencies*</u> or <u>*inefficiencies*</u> of supporting departments

When should <u>actual</u> & <u>budgeted</u> costs be used?

Actual costs should be used for *performance evaluation*.
Budgeted costs should be used for *product costing*.

LEARNING OBJECTIVE

3

Allocate support-department costs to producing departments using the direct, sequential, & reciprocal methods.

MULTIPLE SUPPORT DEPARTMENTS

When a company has multiple support departments that interact, managers must choose an allocation method.

ALLOCATION METHODS:
Multiple Service Departments

- Direct allocation method
  - Allocate support department costs only to producing departments
- Sequential allocation method
  - Allocate support department costs in step-down approach
- Reciprocal allocation method

MULTIPLE SUPPORT DEPARTMENTS: Background

A factory has the following departments

- Producing
  - Grinding
  - Assembly
- Support
  - Power
  - Maintenance

MULTIPLE SUPPORT DEPARTMENTS: Data

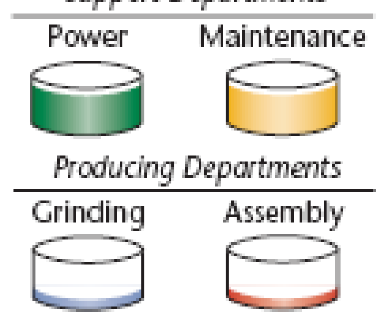
	Support Departments		Producing Departments	
	Power	Maintenance	Grinding	Assembly
Direct costs*	$250,000	$160,000	$100,000	$60,000
Normal activity:				
Kilowatt-hours	—	200,000	600,000	200,000
Maintenance hours	1,000	—	4,500	4,500

*For a producing department, direct costs refer only to overhead costs that are directly traceable to the department.

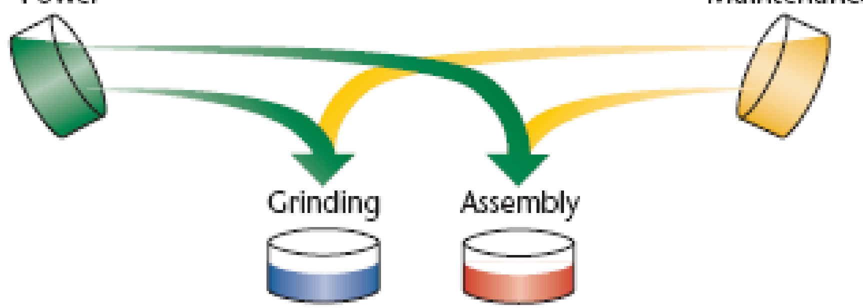
EXHIBIT 7.7

ALLOCATION: Direct Method

Suppose there are two support departments, Power and Maintenance, and two producing departments, Grinding and Assembly, each with a "bucket" of directly traceable overhead cost.
Objective: Distribute all maintenance and power costs to Grinding and Assembly using the direct method.



Direct method—Allocate maintenance and power costs only to Grinding and Assembly.



After allocation—Zero cost in maintenance and power; all overhead cost is in Grinding and Assembly.

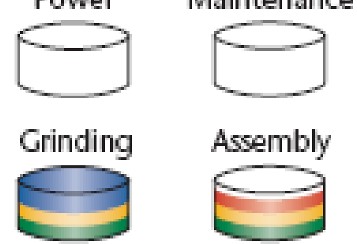


EXHIBIT 7.6

ALLOCATION: Direct Method Step 1

Develop allocation ratios for support department costs.

Step 1—Calculate Allocation Ratios	Grinding	Assembly
Power: $\frac{600{,}000}{(600{,}000 + 200{,}000)}$	0.75	—
$\frac{200{,}000}{(600{,}000 + 200{,}000)}$	—	0.25
Maintenance: $\frac{4{,}500}{(4{,}500 + 4{,}500)}$	0.50	—
Maintenance: $\frac{4{,}500}{(4{,}500 + 4{,}500)}$	—	0.50

EXHIBIT 7.8

ALLOCATION: Direct Method Step 2

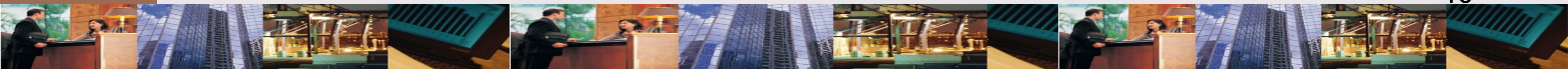
Prorate support department costs to producing depts.

Step 2—Allocate Support-Department Costs Using the Allocation Ratios

	Support Departments		Producing Departments	
	Power	Maintenance	Grinding	Assembly
Direct costs	$ 250,000	$ 160,000	$100,000	$ 60,000
Power[a]	(250,000)	—	187,500	62,500
Maintenance[b]	—	(160,000)	80,000	80,000
	$ 0	$ 0	$367,500	$202,500

[a]Allocation of power based on allocation ratios from Step 1: 0.75 × 250,000; 0.25 × $250,000.
[b]Allocation of maintenance based on allocation ratios from Step 1: 0.50 × $160,000; 0.50 × $160,000.

EXHIBIT 7.8



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ALLOCATION: Sequential Method

Suppose there are two support departments, Power and Maintenance, and two producing departments, Grinding and Assembly, each with a "bucket" of directly traceable overhead cost.

Objective: Distribute all maintenance and power costs to Grinding and Assembly using the sequential method.

Step 1: Rank service departments—#1 Power, #2 Maintenance.

Step 2: Distribute power to Maintenance, Grinding, and Assembly.

Then, distribute maintenance to Grinding and Assembly.

After allocation—Zero cost in Maintenance and Power; all overhead cost is in Grinding and Assembly.

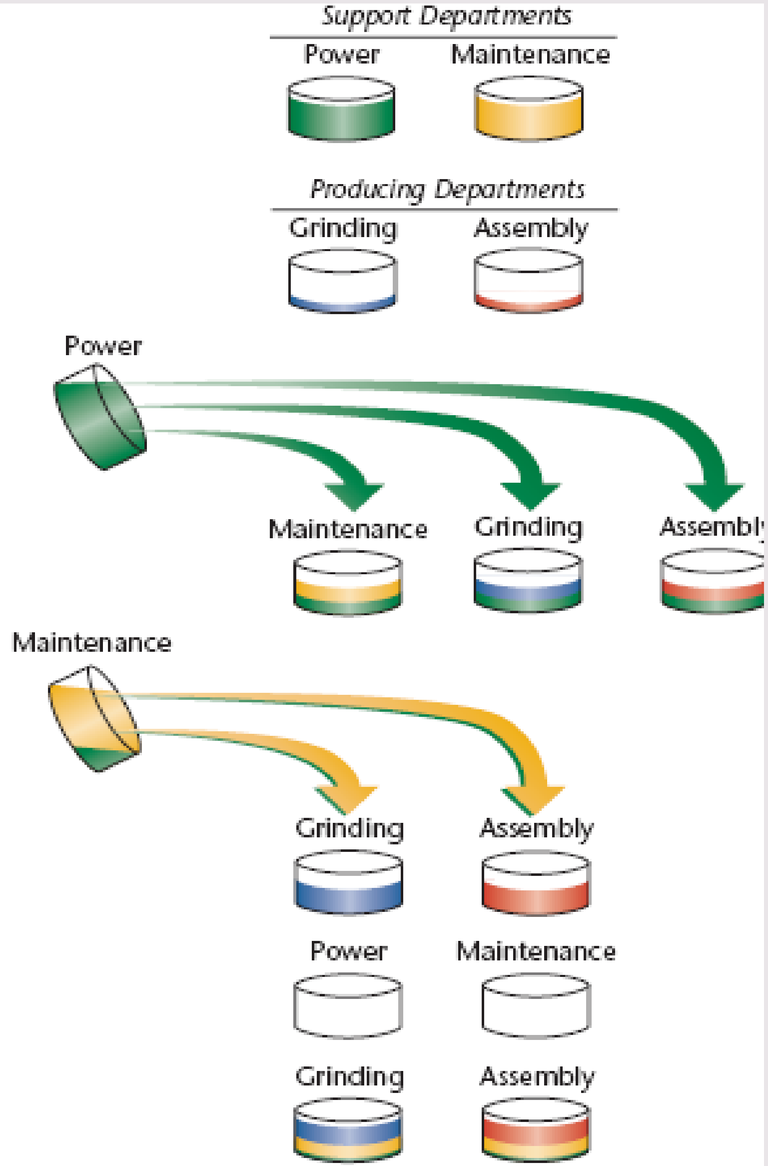

EXHIBIT 7.9

ALLOCATION: Sequential Method
Step 1

Develop allocation ratios for support depts. costs according to ranking.

Step 1—Calculate Allocation Ratios

		Maintenance	Grinding	Assembly
Power:	$\frac{200{,}000}{(200{,}000 + 600{,}000 + 200{,}000)}$	0.20	—	—
	$\frac{600{,}000}{(200{,}000 + 600{,}000 + 200{,}000)}$	—	0.60	—
	$\frac{200{,}000}{(200{,}000 + 600{,}000 + 200{,}000)}$	—	—	0.20
Maintenance:	$\frac{4{,}500}{(4{,}500 + 4{,}500)}$	—	0.50	—
Maintenance:	$\frac{4{,}500}{(4{,}500 + 4{,}500)}$	—	—	0.50

EXHIBIT 7.10

ALLOCATION: Sequential Method
Step 2

Allocate support depts. costs to other departments in order of rankings.

Step 2—Allocate Support-Department Costs Using the Allocation Ratios

	Support Departments		Producing Departments	
	Power	Maintenance	Grinding	Assembly
Direct costs	$ 250,000	$ 160,000	$100,000	$ 60,000
Power[a]	(250,000)	50,000	150,000	50,000
Maintenance[b]	—	(210,000)	105,000	105,000
	$ 0	$ 0	$355,000	$215,000

[a]Allocation of power based on allocation ratios from Step 1: 0.20 × $250,000; 0.60 × $250,000; 0.20 × $250,000.
[b]Allocation of maintenance costs based on allocation ratios from Step 1: 0.50 × $210,000; 0.50 × $210,000.

EXHIBIT 7.10

FORMULAS: Multiple Charging Rates

Allocate each supporting department's costs to ***all*** *other departments before allocating supporting departments' costs to producing departments.*

Allocating Power & Maintenance charges:

P = Direct costs + Share of M. costs

M = Direct costs + Share of P. costs

ALLOCATION: Reciprocal Method
Step 1

Develop allocation ratios for support departments costs.

	Support Departments		Producing Departments	
	Power	Maintenance	Grinding	Assembly
Direct costs:*	$250	$160	$100,000	$60,000
Normal activity:				
Kilowatt-hours	—	200,000	600,000	200,000
Maintenance hours	1,000	—	4,500	4,500
Proportion of Output Used by Department				
	Power	Maintenance	Grinding	Assembly
Allocation ratios:				
Power	—	0.20	0.60	0.20
Maintenance	0.10	—	0.45	0.45

*For a producing department, direct costs are defined as overhead costs that are directly traceable to the department.

EXHIBIT 7.11

ALLOCATION: Reciprocal Method
Step 2

Allocate support depts. costs to producing departments.

Allocate Support-Department Costs Using the Allocation Ratios (Exhibit 7-11) and the Total Support-Department Costs from Reciprocal Method Equations

	Support Departments		Producing Departments	
	Power	Maintenance	Grinding	Assembly
Direct costs	$ 250,000	$ 160,000	$ 100,000	$ 60,000
Power[a]	(271,429)	54,286	162,857	54,286
Maintenance[b]	21,429	(214,286)	96,429	96,429
	$ 0	$ 0	$359,286	$210,715

[a]Power: 0.20 × $271,429; 0.60 × $271,429; 0.20 × $271,429.
[b]Maintenance: 0.10 × $214,286; 0.45 × $214,286; 0.45 × $214,286.

EXHIBIT 7.12

COMPARING ALLOCATION METHODS

Accountants choose between better allocation & cost benefit of easier method.

	Direct Method		Sequential Method		Reciprocal Method	
	Grinding	Assembly	Grinding	Assembly	Grinding	Assembly
Direct costs	$100,000	$ 60,000	$100,000	$ 60,000	$ 100,000	$ 60,000
Allocated from power	187,500	62,500	150,000	50,000	162,857	54,286
Allocated from maintenance	80,000	80,000	105,000	105,000	96,429	96,429
Total cost	$367,500	$202,500	$355,000	$215,000	$359,286	$210,715

EXHIBIT 7.13

LEARNING OBJECTIVE

4

Compute departmental overhead rates.

COSTING PROCESS

Has following steps

- Identify supporting and producing departments
- Allocate supporting department costs to producing departments
- Allocate overhead to producing departments at predetermined rates

LEARNING OBJECTIVE

5

Describe the allocation of joint costs to products. *(Appendix)*.

JOINT PRODUCTS: Definition

A single process produces 2 or more products up to a “split-off” point.

SPLIT-OFF POINT: Definition

The point at which products become separate & identifiable.

LO 5

ACCOUNTING FOR JOINT PRODUCT COSTS

- 3 methods
  - **Physical units**: joint costs distributed on basis of physical units
  - **Sales-value-at-split-off**: joint costs distributed on basis of sales value at split-off
  - **Net realizable value**: joint costs distributed on basis of hypothetical sales value
  - **By-products**: because insignificant sales value, no joint cost allocation

CHAPTER 7

THE END